

PreACT 8/9 Printable Practice Exam 16

Science • Exam 04

30 questions • 30 minutes

PART A - QUESTIONS

Stimulus 1

Biologists sampled a clear lake at 5 depths on the same afternoon. At each depth, they measured the percentage of surface light remaining and counted algae cells in equal-volume water samples. Algae use light for photosynthesis.

Depth (m)	Surface light remaining (%)	Mean algae cells (thousands/mL)
0	100	120
2	72	190
4	41	230
6	18	140
8	5	40

S4-001

At what depth was the mean algae concentration greatest?

- A) 0 m
- B) 2 m
- C) 4 m
- D) 8 m

S4-002

How many more thousand algae cells per milliliter were measured at 4 m than at 8 m?

- A) 41
- B) 90
- C) 190
- D) 270

S4-003

If the percentage of surface light changed linearly between 4 m and 6 m, what percentage would be estimated at 5 m?

- A) 23.0%
- B) 29.5%
- C) 41.0%
- D) 59.0%

S4-004

Which investigation would best test the effect of light level on algae growth independently of lake depth?

- A) Grow equal starting amounts of algae in identical containers under different light levels while keeping other conditions constant.
- B) Collect different algae species from different depths and give each a different nutrient level.
- C) Measure only lake depth at several locations without counting algae.
- D) Place unequal amounts of algae in containers with different temperatures and light levels.

S4-005

Which conclusion is best supported by the measurements?

- A) Algae concentration was greatest at an intermediate depth rather than where light was highest or lowest.
- B) Algae concentration increased continuously as depth increased.
- C) No algae were present below 6 m.
- D) The percentage of surface light increased with depth.

Stimulus 2

Students used a spring scale to pull the same wooden block horizontally at constant speed. The scale reading gave the pulling force. Each result is the mean of 5 trials.

Experiment 1: The block's total mass was 500 g while the surface beneath it was changed.

Surface	Mean pulling force (N)
Smooth plastic	0.8
Paper	1.2
Felt	1.7
Rubber	2.4

Experiment 2: The rubber surface was used while mass was added to the 500 g block.

Added mass (g)	Mean pulling force (N)
0	2.4
250	3.1
500	3.9
750	4.7

S4-006

In Experiment 1, how much greater was the mean pulling force on rubber than on smooth plastic?

- A) 0.8 N
- B) 1.2 N
- C) 1.6 N
- D) 3.2 N

S4-007

What mean pulling force was measured on rubber when 500 g was added to the block?

- A) 2.4 N
- B) 3.1 N
- C) 3.9 N
- D) 4.7 N

S4-008

Which factor did the students deliberately change in Experiment 1?

- A) Surface beneath the block
- B) Block's total mass
- C) Pulling speed
- D) Mean pulling force

S4-009

Why was the same wooden block used throughout both experiments?

- A) To prevent differences in block material and shape from affecting the force comparisons
- B) To make block identity the dependent variable
- C) To ensure the pulling force was identical in every condition
- D) To cause the surface type and added mass to change together

S4-010

A student predicts that increasing the added mass on rubber will require a greater pulling force at constant speed. How do the Experiment 2 results evaluate this prediction?

- A) They support it because mean pulling force increased from 2.4 N to 4.7 N as added mass increased.
- B) They support it because pulling force stayed at 2.4 N for every mass.
- C) They contradict it because the greatest mass required the least force.
- D) They cannot evaluate it because added mass was not changed.

Stimulus 3

After a rainstorm ended, scientists monitored one small watershed. At the same time each day, they measured stream discharge and the depth from the ground surface to groundwater in a nearby well. A smaller groundwater depth means the water table is higher.

Days after storm	Stream discharge (m ³ /s)	Groundwater depth (m)
0	8.2	4.8
1	11.5	4.5
2	9.6	4.1
3	7.4	3.9
4	6.1	4.0
5	5.5	4.2

S4-011

On which day after the storm was stream discharge greatest?

- A) Day 0
- B) Day 1
- C) Day 3
- D) Day 5

S4-012

How much did the water table rise from Day 0 to Day 3?

- A) 0.3 m
- B) 0.9 m
- C) 3.9 m
- D) 8.7 m

S4-013

If stream discharge changed linearly between Day 3 and Day 4, what discharge would be estimated halfway between those measurements?

- A) 5.75 m³/s
- B) 6.75 m³/s
- C) 7.40 m³/s
- D) 13.50 m³/s

S4-014

Which change would most directly improve confidence that the groundwater pattern represents the entire watershed?

- A) Measure groundwater in several wells distributed across the watershed.
- B) Stop measuring stream discharge after Day 1.
- C) Use a different time of day for every well measurement.
- D) Replace groundwater measurements with air-temperature measurements.

S4-015

Which comparison of the two responses to the storm is supported?

- A) Stream discharge peaked on Day 1, while the water table reached its highest level on Day 3.
- B) Both stream discharge and water-table height peaked on Day 5.

- C) The water table reached its highest level before stream discharge peaked.
D) Neither measurement changed after the storm.

Stimulus 4

Students studied pill bugs using a choice chamber with 2 equal sides connected by an opening. They placed 20 pill bugs in the center and counted the animals on each side after 10 minutes. Each result is the mean of 5 trials. The left-right positions of the conditions were switched between trials.

Experiment 1: Both sides were dark, but one side was dry and the other was moist.

Condition	Mean pill bugs
Dry, dark	3
Moist, dark	17

Experiment 2: Both sides were moist, but one side was illuminated and the other was dark.

Condition	Mean pill bugs
Moist, light	6
Moist, dark	14

S4-016

In Experiment 2, what mean number of pill bugs was counted on the moist, dark side?

- A) 3
- B) 6
- C) 14
- D) 17

S4-017

What was the measured response in both experiments?

- A) The number of pill bugs on each side after 10 minutes
- B) The moisture level selected before each trial
- C) The brightness of the illuminated side
- D) The total length of the choice chamber

S4-018

Why were the left-right positions of the conditions switched between trials?

- A) To reduce the chance that a preference for one physical side would be mistaken for a response to the tested condition
- B) To make the total number of pill bugs change between trials
- C) To keep both sides under different conditions in every experiment
- D) To eliminate the need for multiple trials

S4-019

Which conclusion is supported by the two experiments?

- A) Under the tested conditions, pill bugs preferred moist over dry areas and dark over illuminated areas.
- B) Pill bugs preferred dry areas and bright light in both experiments.
- C) Moisture affected the total number of pill bugs placed in the chamber.
- D) Light and moisture produced identical distributions in every trial.

S4-020

If one chamber side were moist and dark while the other were dry and illuminated, which prediction is most consistent with both experiments?

- A) More pill bugs would gather on the moist, dark side.
- B) More pill bugs would gather on the dry, illuminated side.

- C) Exactly 10 pill bugs would occupy each side in every trial.
- D) No pill bugs would enter either side.

Stimulus 5

After rare heavy rain, a shallow lake forms in a desert basin called a playa and usually disappears within 6 days. The basin has no visible surface outlet. During those days, afternoon temperatures often exceed 30°C, humidity is low, and winds are frequent. A salt crust remains after the water disappears. Fractures are present in the playa floor, and porous rock containing groundwater lies beneath the basin.

Viewpoint 1 — Evaporation model: Most lake water changes into water vapor because the air is hot, dry, and windy. Dissolved salts do not evaporate, so they remain as a crust. The floor fractures have little effect on the water loss.

Viewpoint 2 — Infiltration model: Most lake water flows downward through floor fractures into the porous rock. The salt crust forms because some water evaporates, but downward flow is the main reason the lake disappears.

S4-021

Which material remains in the playa after the lake water disappears?

- A) A salt crust
- B) A permanent ice layer
- C) Meteorite fragments
- D) A forest soil

S4-022

Which new result would most directly support Viewpoint 1 over Viewpoint 2?

- A) An evaporation pan beside the lake loses water at nearly the same rate as the lake.
- B) A tracer placed in the lake appears in groundwater beneath the basin.
- C) Water flows rapidly into open fractures in the playa floor.
- D) Groundwater level rises soon after the lake forms.

S4-023

Which finding would most directly support the mechanism in Viewpoint 2?

- A) A harmless tracer added to the lake is later detected in groundwater below the fractures.
- B) Salt remains when a sample of lake water evaporates in a dish.
- C) Wind speed increases during the afternoon.
- D) Water vapor is detected above the lake surface.

S4-024

The two viewpoints disagree primarily about:

- A) whether most water leaves upward as vapor or downward through fractures.
- B) whether the lake forms after heavy rain.
- C) whether a salt crust remains after the water disappears.
- D) whether the basin is located in a desert.

S4-025

Suppose water beneath a floating cover that blocks evaporation remains at nearly the same depth while nearby uncovered water rapidly becomes shallower. This result would most likely:

- A) support Viewpoint 1 because blocking evaporation greatly reduced water loss.
- B) support Viewpoint 2 because the cover opened more floor fractures.
- C) support both viewpoints equally because both require all water to infiltrate.

D) contradict Viewpoint 1 because covered water did not evaporate quickly.

Stimulus 6

Students released a wheeled cart from rest on a ramp and measured its speed at the bottom with a motion sensor. The ramp surface and travel distance were the same in every trial. Each result is the mean of 4 trials.

Experiment 1: Cart mass was held at 200 g while the vertical release height was changed.

Release height (cm)	Mean speed (m/s)
5	0.62
10	0.88
15	1.08
20	1.24

Experiment 2: Release height was held at 15 cm while cart mass was changed.

Cart mass (g)	Mean speed (m/s)
100	1.09
200	1.08
300	1.07
400	1.08

S4-026

In Experiment 1, what mean speed was measured at a release height of 20 cm?

- A) 0.62 m/s
- B) 0.88 m/s
- C) 1.08 m/s
- D) 1.24 m/s

S4-027

How much did mean speed increase when release height increased from 5 cm to 20 cm?

- A) 0.36 m/s
- B) 0.62 m/s
- C) 1.24 m/s
- D) 1.86 m/s

S4-028

In Experiment 1, which quantity was intentionally varied?

- A) Release height
- B) Cart mass
- C) Ramp surface
- D) Mean speed

S4-029

Why were the same cart wheels and ramp surface used in every trial?

- A) To keep rolling conditions from providing another explanation for speed differences
- B) To make wheel type the independent variable
- C) To guarantee identical speed at every release height
- D) To cause cart mass to increase during each run

S4-030

A student claims that doubling cart mass from 200 g to 400 g doubles the speed at a 15 cm release height. How do the data evaluate this claim?

- A) They support it because 1.08 m/s is twice 0.62 m/s.
- B) They support it because the 400 g cart was released from twice the height.
- C) They contradict it because both masses produced the same mean speed of 1.08 m/s.

D) They cannot evaluate it because Experiment 2 did not vary mass.

PART B - ANSWER KEY + EXPLANATIONS

S4-001 - Correct: C) 4 m

Explanation: The greatest mean concentration was 230 thousand cells/mL at a depth of 4 m.

S4-002 - Correct: C) 190

Explanation: The difference was $230 - 40 = 190$ thousand cells/mL.

S4-003 - Correct: B) 29.5%

Explanation: A depth of 5 m is halfway between 4 m and 6 m, so the estimate is halfway between 41% and 18%, or 29.5%.

S4-004 - Correct: A) Grow equal starting amounts of algae in identical containers under different light levels while keeping other conditions constant.

Explanation: Changing only light level would isolate it as the tested factor affecting algae growth.

S4-005 - Correct: A) Algae concentration was greatest at an intermediate depth rather than where light was highest or lowest.

Explanation: Algae concentration peaked at 230 thousand cells/mL at 4 m, while it was lower at both 0 m and 8 m.

S4-006 - Correct: C) 1.6 N

Explanation: The mean forces were 2.4 N on rubber and 0.8 N on smooth plastic, a difference of 1.6 N.

S4-007 - Correct: C) 3.9 N

Explanation: Experiment 2 lists a mean pulling force of 3.9 N for 500 g of added mass.

S4-008 - Correct: A) Surface beneath the block

Explanation: Experiment 1 changed the surface while holding the block's total mass at 500 g.

S4-009 - Correct: A) To prevent differences in block material and shape from affecting the force comparisons

Explanation: Using the same block controlled its material and shape while surface or added mass was tested.

S4-010 - Correct: A) They support it because mean pulling force increased from 2.4 N to 4.7 N as added mass increased.

Explanation: The force increased at every step as added mass rose from 0 g to 750 g.

S4-011 - Correct: B) Day 1

Explanation: The greatest stream discharge was 11.5 m³/s on Day 1.

S4-012 - Correct: B) 0.9 m

Explanation: Groundwater depth decreased from 4.8 m to 3.9 m, meaning the water table rose by 0.9 m.

S4-013 - Correct: B) 6.75 m³/s

Explanation: The halfway value between 7.4 m³/s and 6.1 m³/s is 6.75 m³/s.

S4-014 - Correct: A) Measure groundwater in several wells distributed across the watershed.

Explanation: Several distributed wells would show whether the pattern occurs beyond the single monitored location.

S4-015 - Correct: A) Stream discharge peaked on Day 1, while the water table reached its highest level on Day 3.

Explanation: Discharge was greatest on Day 1, and groundwater depth was smallest—indicating the highest water table—on Day 3.

S4-016 - Correct: C) 14

Explanation: Experiment 2 reports a mean of 14 pill bugs on the moist, dark side.

S4-017 - Correct: A) The number of pill bugs on each side after 10 minutes

Explanation: The students measured how the 20 pill bugs were distributed between the two conditions after 10 minutes.

S4-018 - Correct: A) To reduce the chance that a preference for one physical side would be mistaken for a response to the tested condition

Explanation: Switching sides helps separate responses to moisture or light from a simple left-side or right-side preference.

S4-019 - Correct: A) Under the tested conditions, pill bugs preferred moist over dry areas and dark over illuminated areas.

Explanation: Seventeen of 20 chose moist over dry in Experiment 1, and 14 of 20 chose dark over light in Experiment 2.

S4-020 - Correct: A) More pill bugs would gather on the moist, dark side.

Explanation: The moist, dark side combines the moisture and light conditions that were preferred in the separate experiments.

S4-021 - Correct: A) A salt crust

Explanation: The observations state that a salt crust remains after the water disappears.

S4-022 - Correct: A) An evaporation pan beside the lake loses water at nearly the same rate as the lake.

Explanation: A matching loss rate from an evaporation pan would show that evaporation alone can account for the lake's decline.

S4-023 - Correct: A) A harmless tracer added to the lake is later detected in groundwater below the fractures.

Explanation: Tracer movement from the lake into groundwater would directly demonstrate downward infiltration through the basin floor.

S4-024 - Correct: A) whether most water leaves upward as vapor or downward through fractures.

Explanation: Viewpoint 1 identifies evaporation as the main pathway, whereas Viewpoint 2 identifies infiltration.

S4-025 - Correct: A) support Viewpoint 1 because blocking evaporation greatly reduced water loss.

Explanation: If stopping evaporation also stops most depth loss, evaporation is likely the dominant pathway.

S4-026 - Correct: D) 1.24 m/s

Explanation: The 20 cm row in Experiment 1 lists a mean speed of 1.24 m/s.

S4-027 - Correct: B) 0.62 m/s

Explanation: Mean speed increased from 0.62 m/s to 1.24 m/s, a difference of 0.62 m/s.

S4-028 - Correct: A) Release height

Explanation: Experiment 1 changed release height while holding cart mass at 200 g.

S4-029 - Correct: A) To keep rolling conditions from providing another explanation for speed differences

Explanation: Constant wheels and ramp surface help isolate release height or cart mass as the tested factor.

S4-030 - Correct: C) They contradict it because both masses produced the same mean speed of 1.08 m/s.

Explanation: At the same 15 cm height, the 200 g and 400 g carts both had a mean speed of 1.08 m/s.